



Security, Redundancy & the Spreadsheet

Who can reach what, how the data is protected, and how to get it back.

Everything lives in one Google account, **questwsottawa@gmail.com**. That account owns the spreadsheet, the Drive folder, the script and the web app. Protecting that account protects the whole system — it is the single most important control here, and it is worth having two-factor authentication on it.

Who can reach what

Thing	Who can open it
Quote page	Anyone with the link. It is a public page — that is the point.
Terms and Privacy pages	Anyone. Reachable without submitting anything.
Staff console	Anyone can load the page, but it shows nothing without a valid PIN.
A specific quote	Needs the quote number and the matching last name.
The spreadsheet	Only the Google account. Staff never touch it directly.
Quote PDFs	Anyone with the direct link. Links are not published.
Photos and signed contracts	Anyone with the direct link — deliberately, so staff can open them without a Google login.

The known trade-off, stated plainly

Photo folders and signed contracts are set to "anyone with the link can view". That is what lets a technician open them on a personal phone with no Google account. It also means anyone who obtains a link can view that file. This was an accepted trade-off; locking them to the Google account is a small change if you ever want it.

Staff access

There are no usernames or passwords. A 4-digit PIN identifies the person, which is why the system will not issue a duplicate PIN — if two people shared one, the system would sign the second person in as the first.

Control	How it behaves
Session length	Twelve hours, then a fresh sign-in.
Wrong PINs	Ten in a row pauses login for 15 minutes and emails an alert.
Permissions	view, pay, adjust, email, photos, plus admin. You only see the controls you can use.
Removing someone	Their PIN stops working immediately and every device they are signed in on is ended.
Lockout protection	The console refuses to remove or demote the last admin, or to let you remove your own account.

Control	How it behaves
Activity Log	Every console action — who, what, when — on its own tab in the spreadsheet.

Why PINs rather than Google logins: the account is ordinary Gmail, not Google Workspace, so there is no way to restrict a web app to a list of staff accounts. PINs are the workable alternative.

Protecting customers

- Looking up a quote needs the number **and** the last name. A number alone is not enough.
- The "do you already have a quote?" check reads only the leads list and requires both email and last name. Leads hold no pricing, so the most it can ever reveal is that an address started a quote — it can never be used to pull someone's priced quote.
- People who only reached the pricing and stopped are excluded from every customer email sweep. Without that, a stranger — possibly a competitor — would get a "your quote is waiting" email ten days later.
- The Terms version each customer agreed to, and the timestamp, are stored with their quote rather than only in their browser.

Redundancy — four independent layers

Layer	What it protects against	Where it lives
Nightly backup	Anything that damages the spreadsheet.	A full .xlsx emailed to Chris every evening at 6pm. Keep several.
Google's own history	Accidental edits and deletions.	File → Version history in the spreadsheet, going back automatically.
Pre-write snapshot	A bulk change that makes things worse.	The whole sheet is saved to Drive automatically before a restore or a season re-price writes anything, and the link comes back with the result.
Backend version history	A bad software change.	Every deploy creates a numbered version. Rolling back takes about ten seconds.

Restoring from a backup

Admin only. Console → menu → **Restore from backup**.

- Save the .xlsx from one of the nightly emails — pick the last one from before the problem.
- Upload it. The console reads it and shows what is missing from the sheet, what differs, and what exists only on the sheet. **Nothing has been written at this point.**
- Choose: put back only what is missing (the safe option), or also overwrite quotes that differ.
- Confirm. It reports how many were restored and links the snapshot it saved first.

Two guarantees: **a restore never deletes a live quote** — anything taken since the backup survives either option — and the current sheet is always saved to Drive first, so the restore itself can be undone by restoring that snapshot.

The spreadsheet

Named **Winter Quotes 2026-2027**. One tab per storage location, plus supporting tabs. A quote lives on exactly one tab, and moves automatically if its storage changes.

Tab	What is on it
Inside	Quotes stored inside.
Premium Inside	Quotes in premium inside storage.
Outside	Quotes stored outside.
No Storage	Services only — the customer keeps the unit.
Golf Cart / E-Bike	One tab each; these units have no storage choice.
Quote Started	People who reached the pricing and stopped. Excluded from every customer email sweep.
Activity Log	Every console action, with who and when.

What each row holds

Twenty-three columns: name, quote number, balance, timestamp, status, unit, phone, email, year/make/model, dimensions, total, deposit, pay choice, itemised services, quotes requested, customer notes, PDF link, sign link, reminder marker, **payload**, paid, and photos.

The **Payload** column is the important one. It holds the complete quote as structured data — every selection the customer made, every staff change, the payment ledger, the email history, the Terms they agreed to. It is what makes reloading, re-pricing, correcting and next year's rollover possible. It looks like unreadable text in the cell; that is expected.

Do not hand-edit money or email cells

Editing a total, a payment or a customer email directly in the spreadsheet changes the cell but not the underlying record, so the sheet, the PDF and the customer's own page will disagree from that moment on. Use the console or the spreadsheet menu, which update everything together. Reading the sheet is always safe.

Why the customer's selections are stored, not just the prices

Because a quote is a description of a job, not a number. Storing the selections means an old quote can be re-priced against this year's rates on reload, a corrected measurement can flow through every affected line, and next season can start from what each customer had last season.

One field on a quote is deliberately ours only

The staff note — why a quote is the way it is — is never sent to a customer. It is not on the PDF, not in any email, and not returned when a customer reloads their quote on the public page. Anyone with the console permission to read notes can read all of them, so write them as working notes about the job rather than about the person.

How the software gets updated

The customer page, Terms, Privacy and the console are static files — they go live as soon as they change. The backend is deployed deliberately, from a button, and every deploy:

- runs the full automated check suite first and **refuses to deploy if anything fails**;
- creates a numbered, permanent version;
- points the existing web address at it, so the address customers and staff use never changes.

Rolling back is choosing the previous version from a dropdown — about ten seconds. The check suite covers, among other things: that prices have not moved unintentionally, that the customer page and backend agree on every rule, that people who only started a quote are never emailed, that credits are never hidden, that a re-measure survives a customer save, that a backup restore always snapshots first, and that the last admin cannot be removed.

Where the real risks are

Risk	What reduces it	What is left
The Google account is compromised	Strong password; two-factor authentication.	This is the big one. Everything is in that account.
A PIN is shared or seen	Per-person PINs, 12-hour sessions, activity logging, per-permission limits.	Reset the PIN from the console; it takes effect at once.
A file link is forwarded	Links are not published anywhere.	Anyone holding a photo or contract link can open it.
A bad software change	Automated checks gate the deploy; versions roll back in seconds.	Front-end pages go live immediately on change.
Spreadsheet damaged or deleted	Nightly backups, Google version history, restore tool.	Up to one day of work if the sheet is lost outright.
Google outage	None available.	The system is unavailable until Google recovers. Quotes can be taken on paper.

If something goes badly wrong

Situation	First move
A quote looks wrong after an update	Roll the backend back one version, then tell Claude what changed.
Quotes are missing from the sheet	Restore from last night's backup, missing-only mode. Nothing live is deleted.
Someone edited cells by hand	Spreadsheet → File → Version history to undo, or restore the quote from a backup.
A staff member leaves	Console → Staff & permissions → Remove. Signs them out everywhere immediately.
Suspected account compromise	Change the Google password and turn on two-factor first. Then reset every staff PIN from the console.